MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 06 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

A number is selected at random from the numbers 1 to 25. The probability that the selected number is prime is:

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$

(c) $\frac{10}{25}$ (d) $\frac{7}{25}$

Working & Calculation:

- Sample space: $S = \{1, 2, 3, \ldots, 25\} \implies n(S) = 25$.
- Prime numbers in this range are $E = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ (9 primes):

$$n(E) = 9 \implies P(\text{prime number}) = \frac{n(E)}{n(S)} = \frac{9}{25}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{9}{25}$

Question 2 [1 Mark]

The probability that a non-leap year contains 53 Mondays is:

(a) $\frac{1}{7}$ (b) $\frac{2}{7}$

(c) $\frac{5}{7}$ (d) 0

Working & Calculation:

- A ordinary (non-leap) year contains 365 days = 52 weeks + 1 extra day.
- The 52 weeks contain exactly 52 Mondays.
- The 53rd Monday occurs if and only if the single extra day falls on a Monday.
- Sample space for the extra day: $S = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\} \implies n(S) = 7$.

$$P(\text{53 Mondays in a non-leap year}) = \frac{1}{7}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{7}$

Question 3 [1 Mark]

A single letter is selected at random from the word 'PROBABILITY'. The probability that it is a vowel is:

(a) $\frac{3}{11}$ (b) $\frac{4}{11}$

(c) $\frac{5}{11}$ (d) $\frac{2}{11}$

Working & Calculation:

- Total letters in 'PROBABILITY': $n(S) = 11$.
- The vowels present are {O, A, I, I} $\implies$ $n(\text{vowels}) = 4$.

$$P(\text{a vowel}) = \frac{n(\text{vowels})}{n(S)} = \frac{4}{11}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow$ $\frac{4}{11}$

Question 4 [1 Mark]

A number x is chosen at random from the set $\{-3, -2, -1, 0, 1, 2, 3\}$. The probability that $|x| < 2$ is:

(a) $\frac{2}{7}$ (b) $\frac{3}{7}$

(c) $\frac{4}{7}$ (d) $\frac{5}{7}$

Working & Calculation:

- Given set $S = \{-3, -2, -1, 0, 1, 2, 3\}$ $\implies$ $n(S) = 7$.
- The condition $|x| < 2$ means $-2 < x < 2$.
- Elements of S satisfying this inequality are $E = \{-1, 0, 1\}$ $\implies$ $n(E) = 3$.

$$P(|x| < 2) = \frac{n(E)}{n(S)} = \frac{3}{7}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow$ $\frac{3}{7}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): A lot contains 12 defective pens and 132 good pens. The probability that a pen taken out at random is good is $\frac{11}{12}$.

Reason (R): Total pens $= 12 + 132 = 144$, and favourable outcomes for a good pen $= 132$ $\implies$ $\frac{132}{144} = \frac{11}{12}$.

Options:

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**
(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**
(c) **(A) is true, but (R) is false.**
(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Reason (R):** Total pens $= 12 + 132 = 144$. Favourable outcomes $= 132$. The ratio is $\frac{132}{144} = \frac{11}{12}$. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** Probability of drawing a good pen is indeed $\frac{132}{144} = \frac{11}{12}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** (R) contains the exact calculation establishing (A). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish. What is the probability that the fish taken out is a male fish?

Sample Space: Total number of fish in the aquarium tank: $n(S) = 5 + 8 = 13$.

Calculation: Number of favourable outcomes (male fish): $n(\text{male}) = 5$.

$$P(\text{male fish}) = \frac{n(\text{male})}{n(S)} = \frac{5}{13}$$

Answer: $\mathbf{P(\text{male fish}) = \frac{5}{13}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability of getting:
(i) a spade or a red king (ii) neither a jack nor a queen (iii) a black face card.

Sample Space: Total cards in a standard deck: $n(S) = 52$.

(i) A spade or a red king [1 Mark]

Number of spades = 13. Red kings are King of Hearts and King of Diamonds (2 cards). Since spades are black, the sets are disjoint: $n(E_1) = 13 + 2 = 15$:

$$P(\text{spade or red king}) = \frac{15}{52}$$

Answer (i): $\mathbf{P(\text{spade or red king}) = \frac{15}{52}}$

(ii) Neither a jack nor a queen [1 Mark]

There are 4 jacks and 4 queens (8 cards total). Cards that are neither jack nor queen: $52 - 8 = 44$:

$$P(\text{neither jack nor queen}) = \frac{44}{52} = \frac{11}{13}$$

Answer (ii): $\mathbf{P(\text{neither jack nor queen}) = \frac{11}{13}}$

(iii) A black face card [1 Mark]

Black face cards consist of the Jack, Queen, and King of Spades and Clubs ($2 \times 3 = 6$ cards):

$$P(\text{black face card}) = \frac{6}{52} = \frac{3}{26}$$

Answer (iii): $\mathbf{P(\text{black face card}) = \frac{3}{26}}$

Question 8 [3 Marks]

A box contains cards numbered 3, 5, 7, 9, . . . , 35, 37. A card is drawn at random from the box. Find the probability that the number on the drawn card is:
(i) a prime number (ii) a number divisible by 7 (iii) a two-digit number.

Card Count: The cards form an arithmetic sequence of odd numbers $3, 5, 7, \ldots, 37$:

$$n(S) = \frac{37-3}{2} + 1 = \frac{34}{2} + 1 = 18 \text{ cards}$$

(i) A prime number [1 Mark]

Odd prime numbers in $\{3, 5, \ldots, 37\}$ are $E_1 = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\} \implies n(E_1) = 11$:

$$P(\text{prime number}) = \frac{11}{18}$$

Answer (i): P(prime number) = $\frac{11}{18}$

(ii) A number divisible by 7 [1 Mark]

Odd multiples of 7 in the set are $E_2 = \{7, 21, 35\} \implies n(E_2) = 3$:

$$P(\text{divisible by } 7) = \frac{3}{18} = \frac{1}{6}$$

Answer (ii): P(divisible by 7) = $\frac{1}{6}$

(iii) A two-digit number [1 Mark]

Single-digit cards are $\{3, 5, 7, 9\}$ (4 cards). Remaining cards are two-digit: $18 - 4 = 14$:

$$P(\text{two-digit number}) = \frac{14}{18} = \frac{7}{9}$$

Answer (iii): P(two-digit number) = $\frac{7}{9}$

Question 9 [3 Marks]

A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random from the lot.
(i) What is the probability that this bulb is defective?
(ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective?

Given Setup: Total bulbs = 20, defective bulbs = 4, non-defective bulbs = $20 - 4 = 16$.

(i) Probability that the bulb drawn is defective [1 Mark]

Favourable defective bulbs = 4:

$$P(\text{defective}) = \frac{4}{20} = \frac{1}{5}$$

Answer (i): P(defective) = $\frac{1}{5}$

(ii) Probability that second bulb drawn without replacement is not defective [2 Marks]

When a non-defective bulb is drawn and not replaced:

- New total number of bulbs: $n(S') = 20 - 1 = 19$.
- Remaining non-defective bulbs: $16 - 1 = 15$.

$$P'(\text{not defective}) = \frac{15}{19}$$

Answer (ii): P(not defective second time) = $\frac{15}{19}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10 (Case Study Analysis) [4 Marks]

An education department conducted an audit across 100 secondary schools to analyze computer-based learning facilities. The survey categorized schools based on the number of working computers available for students:

Computer Range	1–10	11–20	21–50	51–100	More than 100
Number of Schools	15	25	35	18	7

A school is selected at random from the surveyed group. Based on the above scenario, answer the following:

Total Surveyed Population: $n(S) = 15 + 25 + 35 + 18 + 7 = 100$ schools.

(i) Probability that the selected school has more than 50 computers [1 Mark]

Schools with more than 50 computers: (51–100) + (More than 100) $= 18 + 7 = 25$:

$$P(\text{more than } 50) = \frac{25}{100} = \frac{1}{4} = 0.25$$

Answer (i): $\mathbf{P(\text{more than } 50) = \frac{1}{4} = 0.25}$

(ii) Probability that the selected school has at most 20 computers [1 Mark]

Schools with at most 20 computers: (1–10) + (11–20) $= 15 + 25 = 40$:

$$P(\text{at most } 20) = \frac{40}{100} = \frac{2}{5} = 0.4$$

Answer (ii): $\mathbf{P(\text{at most } 20) = \frac{2}{5} = 0.4}$

(iii) Probability that the school has between 11 and 50 computers (inclusive) [1 Mark]

Schools in this range: (11–20) + (21–50) $= 25 + 35 = 60$:

$$P(11 \le \text{computers} \le 50) = \frac{60}{100} = \frac{3}{5} = 0.6$$

Answer (iii): $\mathbf{P(11 \le \text{computers} \le 50) = \frac{3}{5} = 0.6}$

(iv) Probability that the selected school has at least 21 computers [1 Mark]

Schools with at least 21 computers: (21–50) + (51–100) + (More than 100) $= 35 + 18 + 7 = 60$:

$$P(\text{at least } 21) = \frac{60}{100} = \frac{3}{5} = 0.6 \quad \left[\text{or via complement: } 1 - \frac{40}{100} = \frac{60}{100} = 0.6\right]$$

Answer (iv): $\mathbf{P(\text{at least } 21) = \frac{3}{5} = 0.6}$

— END OF WORKED SOLUTIONS (TEST–06) · MATHS MANTHAN TUITION CLASSES —